Scientific Justification *Be sure to include overall significance to astronomy. For standard proposals limit text to one page with figures, captions and references on no more than two additional pages.*

Abstract

We propose near-infrared (NIR) imaging of the recoiling active galactic nucleus (AGN) candidate CID-42. We have selected this candidate based on having offset broad emission lines and existing radio detections. Our goal is to measure the offset between the AGN and the photometric center of its host galaxy. We will model the host galaxy light distribution using parametric fitting with GALFIT to determine its center, and compare this to the AGN position constrained by high-resolution radio observations. A statistically significant spatial offset between the AGN and host galaxy center would provide direct evidence for a recoiling black hole. Confirming such systems would allow us to improve our understanding of AGN kinematics, constrain black hole merger dynamics, and provide observational inputs for future gravitational wave missions such as LISA and pulsar timing arrays.

Binary black hole (BBH) mergers are a fundamental consequence of hierarchical galaxy evolution. Through ground-based gravitational wave (GW) observatories (e.g., LIGO) and pulsar timing arrays (PTAs), many of these mergers have been observed through GW detections [add citation here!]. Numerical relativity simulations predict that the coalescence of a BBH system can impart a recoil velocity, or "kick", to the remnant black hole (BH) due to anisotropic GW emission [Campanelli, 2007]. These kicks can reach velocities of up to 4000 kms $^{-1}$, sufficient to displace the black hole from the center of its host galaxy. Despite strong theoretical motivation, no recoiling, or displaced, active galactic nucleus (AGN) has been confirmed.

Spectroscopic surveys have identified many recoiling AGN candidates through velocity offsets in their broad emission lines; however, these signatures are highly debated and cannot conclusively identify the nature of the system on their own. Other alternative scenarios that can explain these offset emission lines include supermassive black hole binaries (SMBHBs) or spiral arms. Long-term spectroscopic monitoring can distinguish these cases by tracking velocity offsets through time, but such campaigns require observations over extended timescales and are observationally expensive. As a result, recent studies have used high-resolution photometric observations to distinguish these systems from alternate scenarios. For example, CID-42, one of the most compelling recoiling AGN candidates, has been established to be a pair of merging galaxies [Li et. al, 2024].

Consequently, we propose near-infrared (NIR) imaging of CID-42's host galaxy to measure spatial offsets between its photometric center and the AGN. A recoiling AGN is expected to appear as an off-nuclear AGN relative to a well-defined host galaxy center. In contrast, binary SMBHBs or spiral arms will not necessarily produce a significant spatial displacement from the galaxy's center. To determine the host galaxy's center, we will model the light profile using parametric surface brightness fitting as implemented in GALFIT. NIR observations are critical to produce a smooth light profile that robustly characterizes the galaxy's stellar population. The AGN position will be constrained through high-resolution radio observations. CID-42 was chosen based on 2 main criteria: (1) significant velocity offsets in its broad emission lines and (2) available radio detections. By comparing the NIR-constrained host galaxy center to the radio-constrained AGN location, we can assess the statistical significance of the spatial offsets. A statistically significant offset would provide strong evidence for a recoiling AGN. This would offer a rare observational probe of BH merger dynamics and enable studies of AGN feedback in off-nuclear environments.

Even in the absence of a significant offset, this approach will yield important constraints. We will be able to place limits on the reliability of spectroscopic selection techniques for recoiling AGN and help distinguish between competing interpretations such as SMBHBs or complex gas kinematics. In either outcome, this program addresses a fundamental gap between theoretical predictions and observational evidence for recoiling AGN.

NOTE: STILL NEED TO ADD FIGURES AND MAKE REFERENCES SECTION!! Figure below is part of the template, but will be changed in the final draft!

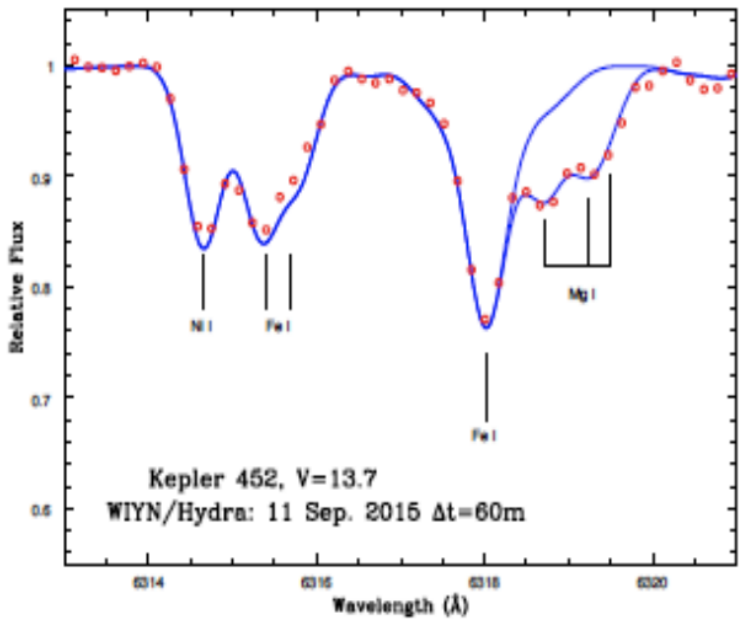

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Figure 1: This is an example of embedded PDF figure

Experimental Design *Describe your overall observational program. How will these observations contribute toward the accomplishment of the goals outlined in the science justification?*

1. The Proposed Target

The proposed target of this proposal is the $z = 0.359$ CID-42 galaxy. Originally, Comerford et al. (2009) thought this was a dual AGN due to the two brightness peaks resolved using Hubble Space Telescope's optical imaging. It was later reinterpreted by Civano et al. (2010) as a recoiling AGN candidate based on offset broad emission lines. Subsequently, Novak et al. (2015) constrained one core using radio observations from the 3GHz Karl G. Jansky Very Large Array (VLA). They found that the entirety of the radio emission came from the southeastern(SE)core, so this is likely associated with the AGN. The northwestern (NW) core did not show radio emissions; thus, ruling out the dual AGN scenario.

The SE core is located at coordinates $\alpha_{J2000} = 10^h00^m43.^{15s}$ and $\delta_{J2000} = +02°06'37.19''$ with a positional error of 0.03".

We will determine whether the NW component represents the photometric center of the host galaxy to test the recoiling AGN scenario. This requires constraining the center of the host galaxy light distribution to a precision comparable to the radio emission positional uncertainty. Therefore, we aim to measure the host galaxy center to an accuracy of 0.03" using NIR imaging. This level of precision will enable a direct comparison between the host galaxy's center and the radio-constrained AGN position, allowing us to assess the statistical significance of their spatial offset.

Candidate Selection Criteria

CID-42 was selected based on the following criteria:
1. Velocity offsets in broad emission lines: This provides spectroscopic evidence for a recoiling AGN
2. Existing high-resolution radio detections: This constrains the AGN position relative to the host
3. Bright host galaxy light profile: This allows us to photometrically constrain the host galaxy center using parametric fitting via GALFIT

2. Signal-to-Nose Ratio Requirement

Our primary goal is to measure the statistical significance of the spatial offset between the AGN and the host galaxy's center. This requires precise measurements of the host galaxy's light profile. We will model the light profile using GALFIT, which determines the center by fitting to a Sersic profile.

The uncertainty on the galaxy's center can be written as:
$\sigma_{center} = \frac{FWHM}{S/N}$,
$S/N = \frac{FWHM}{\sigma_{center}}$
As explained above, we aim to constrain our NW core to have a positional uncertainty (σ_{center}) of 0.03". Assuming NIR seeing at Gemini is FWHM=0.6, we require a minimum S/N = 20. However, this is the minimum requirement for a point source. Since we want to obtain a robust light distribution fit with GALFIT for the entire galaxy, we will adopt a more conservative requirement of S/N = 50.

3. Instrument Choice and Exposure Time Calculation

We propose to use Gemini with FLAMINGOS-2 due to its field of view (FOV) and resolution, NIR capabilities, and exposure time considerations. FLAMINGOS-2 has a sufficiently wide FOV (6.1' x 6.1'), which is necessary to capture the AGN's host galaxy and surrounding stars in a single frame. Its wide-field mode 0.18"/pixel will also allow us to precisely constrain the center of the host galaxy. We will need observations in the NIR (J, H, and K bands) to obtain a smooth light profile of the host galaxy and minimize clumpy star-forming regions. Since the AGN will be unresolved in the NIR, we will constrain its location using radio observations. We can crossmatch our observations and then get an accurate reading of the AGN offset from its host.

Exposure times were estimated using the Gemini FLAMINGOS-2 Integration Time Calculator with a target S/N = 50 for each of the J, H, and K bands. In order to avoid saturating the detector of the galaxy's core and achieve our target S/N, we plan to obtain 5 exposures of 3 seconds in each band. We request a total telescope time of 35 minutes to capture these observations. Exposure time calculations are shown below.

Gemini Integration Time Calculator
Flamingos2 - 2026B.1.1.1

Click here for help with the results page.
software aperture diameter = 0.71 arcsec
fraction of source flux in aperture = 0.62
enclosed pixels = 12.15

derived image size for Gaussian source = 0.60

Contributions to total noise (e-) in aperture (per exposure):
Source noise = 110.76
Background noise = 193.38
Dark current noise = 1.91
Readout noise = 17.43

Total noise per exposure = 223.54
Total signal per exposure = 12268.15

Intermediate S/N for one exposure = 54.88

S/N for the whole observation = 92.65 (including sky subtraction)

Requested total integration time = 15.00 secs, of which 15.00 secs is on source.

The peak pixel signal + background is 4611 e- (1038 ADU). This is 3% of the full well depth of 155400 e-.

Observation Overheads		
Setup	480.0 s	
Telescope offset	4 x 7.0 s	assuming ABAB dithering pattern
Exposure	5 x 3.0 s	
Readout	5 x 20.0 s	
DHS Write	5 x 10.0 s	
Program time	**11 mins 13 secs**	

Input Parameters:
Instrument: Flamingos 2

Source spatial profile, brightness, and spectral distribution:
The z = 0.35900 Gaussian source with FWHM = 0.6 arcsec is a 18.0 AB elliptical-galaxy in the J band.

Instrument configuration:

Optical Components:
- Filter: J
- Fixed Optics
- Detector - 2048x2048 Hawaii-II (HgCdTe)
- Read Mode: Faint Object
- Read Noise: 5.0 e-

Pixel Size: 0.18

Telescope configuration:
- silver mirror coating.
- side looking port.
- wavefront sensor: oiwfs

Observing Conditions:
- Airmass: 1.50
- Image Quality: 70% (≤ 0.47" at zenith, ≤ 0.60" on-source)
- Cloud cover: 70%
- Water Vapor: 50%
- Sky Background: 50%

Likelihood of execution: 12%

Calculation and analysis methods:
- Mode: imaging
- Calculation of S/N ratio with 5 exposures of 3.00 secs, and 100.00% of them on source.
- Analysis performed for aperture that gives 'optimum' S/N and a sky aperture that is 1.00 times the target aperture.

Gemini Integration Time Calculator
Flamingos2 - 2026B.1.1.1

Click here for help with the results page.
software aperture diameter = 0.71 arcsec
fraction of source flux in aperture = 0.62
enclosed pixels = 12.15

derived image size for Gaussian source = 0.60

Contributions to total noise (e-) in aperture (per exposure):
Source noise = 184.29
Background noise = 614.86
Dark current noise = 1.91
Readout noise = 17.43

Total noise per exposure = 642.12
Total signal per exposure = 33963.39

Intermediate S/N for one exposure = 52.89

S/N for the whole observation = 85.41 (including sky subtraction)

Requested total integration time = 15.00 secs, of which 15.00 secs is on source.

The peak pixel signal + background is 35357 e- (7963 ADU). This is 23% of the full well depth of 155400 e-.

Observation Overheads		
Setup	480.0 s	
Telescope offset	4 x 7.0 s	assuming ABAB dithering pattern
Exposure	5 x 3.0 s	
Readout	5 x 20.0 s	
DHS Write	5 x 10.0 s	
Program time	**11 mins 13 secs**	

Input Parameters:
Instrument: Flamingos 2

Source spatial profile, brightness, and spectral distribution:
The z = 0.35900 Gaussian source with FWHM = 0.6 arcsec is a 17.5 AB elliptical-galaxy in the H band.

Instrument configuration:

Optical Components:
- Filter: H
- Fixed Optics
- Detector - 2048x2048 Hawaii-II (HgCdTe)
- Read Mode: Faint Object
- Read Noise: 5.0 e-

Pixel Size: 0.18

Telescope configuration:
- silver mirror coating.
- side looking port.
- wavefront sensor: oiwfs

Observing Conditions:
- Airmass: 1.50
- Image Quality: 70% (≤ 0.47" at zenith, ≤ 0.60" on-source)
- Cloud cover: 70%
- Water Vapor: 50%
- Sky Background: 50%

Likelihood of execution: 12%

Calculation and analysis methods:
- Mode: imaging
- Calculation of S/N ratio with 5 exposures of 3.00 secs, and 100.00% of them on source.
- Analysis performed for aperture that gives 'optimum' S/N and a sky aperture that is 1.00 times the target aperture.

Gemini Integration Time Calculator
Flamingos2 - 2026B.1.1.1

Click here for help with the results page.
software aperture diameter = 0.71 arcsec
fraction of source flux in aperture = 0.62
enclosed pixels = 12.15

derived image size for Gaussian source = 0.60

Contributions to total noise (e-) in aperture (per exposure):
Source noise = 251.75
Background noise = 632.77
Dark current noise = 1.91
Readout noise = 17.43

Total noise per exposure = 681.24
Total signal per exposure = 63380.20

Intermediate S/N for one exposure = 93.04

S/N for the whole observation = 152.40 (including sky subtraction)

Requested total integration time = 15.00 secs, of which 15.00 secs is on source.

The peak pixel signal + background is 40872 e- (9205 ADU). This is 26% of the full well depth of 155400 e-.

Observation Overheads		
Setup	480.0 s	
Telescope offset	4 x 7.0 s	assuming ABAB dithering pattern
Exposure	5 x 3.0 s	
Readout	5 x 20.0 s	
DHS Write	5 x 10.0 s	
Program time	**11 mins 13 secs**	

Input Parameters:
Instrument: Flamingos 2

Source spatial profile, brightness, and spectral distribution:
The z = 0.35900 Gaussian source with FWHM = 0.6 arcsec is a 17.0 AB elliptical-galaxy in the K band.

Instrument configuration:

Optical Components:
- Filter: K_LONG
- Fixed Optics
- Detector - 2048x2048 Hawaii-II (HgCdTe)
- Read Mode: Faint Object
- Read Noise: 5.0 e-

Pixel Size: 0.18

Telescope configuration:
- silver mirror coating.
- side looking port.
- wavefront sensor: oiwfs

Observing Conditions:
- Airmass: 1.50
- Image Quality: 70% (≤ 0.47" at zenith, ≤ 0.60" on-source)
- Cloud cover: 70%
- Water Vapor: 50%
- Sky Background: 50%

Likelihood of execution: 12%

Calculation and analysis methods:
- Mode: imaging
- Calculation of S/N ratio with 5 exposures of 3.00 secs, and 100.00% of them on source.
- Analysis performed for aperture that gives 'optimum' S/N and a sky aperture that is 1.00 times the target aperture.